

DEVA ANAND

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Education

MS, Computer Science, University of Massachusetts Amherst 2025–2027

Advanced NLP, Advanced Machine Learning, Optimization in CS, Systems for Data Science, Performance Engineering

BE, Computer Science and Engineering, Vels Institute of Science, Technology and Advanced Studies 2019–2023

Professional Experience

Wysa, Backend Engineer | Bengaluru, India Jul 2023 – Jul 2025

- Engineered a multi-tenant, database-backed NHS eTriage web backend (Node.js, MongoDB, REST) for **8+ UK clinical clients**, capturing **10,000+ monthly self-referral submissions** via Mayden iaptus REST integrations; contributed to **\$340K+** in revenue.
- Architected a full-stack data-collection and annotation platform (React front-end, Node.js/MongoDB back-end) used daily by the AI team, cutting manual annotation by **100+ hours/month** and lifting labeling throughput by **60%**.
- Implemented backend controls for sensitive participant data, **AWS KMS** encryption for PII/clinical records, 90-day retention, auth hardening, and rate limiting, and remediated **20+ VAPT findings**.
- Served as primary technical POC for the eTriage backend; ran knowledge transfer onboarding a second developer to on-call readiness in **~6 weeks** and piloted a COO-sponsored GitHub Copilot AI-assisted-development experiment.

Cario Growth Services, Backend Developer Intern | Chennai, India Nov 2022 – May 2023

- Designed and shipped **20+ RESTful APIs** (Node.js, Fastify, PostgreSQL) with input validation, clear contracts, and backward-compatible changes used in production by the frontend team.
- Migrated core repositories from JavaScript to TypeScript and shipped a production LLM auto-commenting feature (**4-bit quantized Llama-7B via llama.cpp**) with async queued generation and content-safety filtering.

Freelance / Open Source, Full-Stack Engineer | Remote Jul 2022 – Aug 2023

- Built and deployed **Underdogs Fitness** (Website), a production MERN web platform with Stripe + manual-cash hybrid payments, cron automation, and multi-branch support; live **3+ years**.
- Delivered a second MERN client engagement (US-based inventory management system) end-to-end as a paid freelance build.

Projects

AgenticSearch – Provenance-First Entity Discovery (GitHub | Demo) Python, FastAPI, OpenAI / Groq, SQLite

- Built a Python/FastAPI service turning open-ended queries into structured entity tables via typed retrieval planning, reranking, and deterministic extraction with LLM-API fallback and **cell-level provenance**; hardened with **150+ tests**.

Sonare – Offline Sign / Speech App (GitHub) React, Electron, FastAPI, MediaPipe, whisper.cpp

- Engineered a React/Electron front-end and FastAPI back-end for an offline sign/speech desktop app with on-device NLP/ASR (MediaPipe, whisper.cpp), designing async APIs to keep the UI responsive under local video and audio streams.

KG2RAG-Enhanced – Multi-Hop QA on HotpotQA (GitHub) Python, Ollama / LLaMA-3, sentence-transformers, RRF

- Built the multi-view retrieval component of a KG-guided RAG pipeline (query-side decomposition, RRF fusion, cross-encoder reranking, MMR), lifting supporting-fact recall **+2.68%** over the KG²RAG baseline on a 4,905-question evaluation.

Publications

- “Alzheimer’s Disease Classification using Transfer Learning,” IEEE CONIT 2023, first author, deep transfer learning on neuroimaging data. (Paper)

Skills

Languages & Databases: Python, SQL, JavaScript, TypeScript, HTML/CSS, C++, Java, PostgreSQL, MongoDB, SQLite

Web & Frameworks: FastAPI, Node.js, Express, React, REST APIs, Docker, Git, GitHub Actions

ML & AI: PyTorch, Hugging Face Transformers, NLP, LLM agents, RAG, OpenAI / Groq APIs, evaluation

Retrieval / Search: RRF, cross-encoder reranking, MMR, sentence-transformers, Brave Search, cell-level provenance

Cloud / Platforms: AWS, GCP, Azure, Linux, MongoDB Atlas

Achievements

- Selected for **GirlScript Summer of Code 2023**; contributed to Linkfree, Freehit, and ProjectsHut. - Tutored **30+ underprivileged students** in programming at Sayur, a non-profit in Tamil Nadu.